

Bellman Equation and Value Function Iteration

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Recap: Markov Decision Process (MDP)

- ▶ An MDP is defined by the tuple

$$(\mathcal{S}, \mathcal{A}, P, r, \beta)$$

where:

- ▶ **State space $\mathcal{S}$** : Possible system states.
- ▶ **Action space $\mathcal{A}(s)$** : Feasible actions when in state s .
- ▶ **Transition kernel $P_t(s' | s, a)$** : Probability of moving from s to s' given action a .
- ▶ **Reward function $r_t(s, a)$** : Instantaneous payoff from taking action a in state s . (Sometimes expressed as a *cost* $-r(s, a)$.)
- ▶ **Discount factor $\beta \in (0, 1)$** : Weights future rewards relative to current ones.
- ▶ **Objective**: Choose a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ to maximize expected discounted rewards:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^T \beta^t r_t(s_t, a_t) + \beta^T R_T(s_T) \right].$$

Value Function

- ▶ $V_t^\pi(s)$: The value function corresponding to a policy π

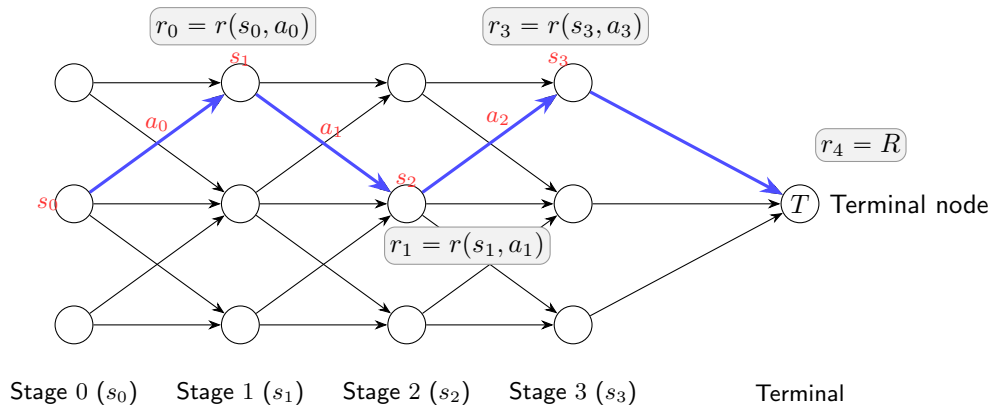
$$V_t^\pi(s) \equiv \mathbb{E}_\pi \left[\sum_{\tau=t}^{T-1} \beta^{\tau-t} r_\tau(s_\tau, a_\tau) + \beta^{T-t} R_T(s_T) \mid s_t = s \right].$$

Note that how the policy $a = \pi(s)$ impact the value value:

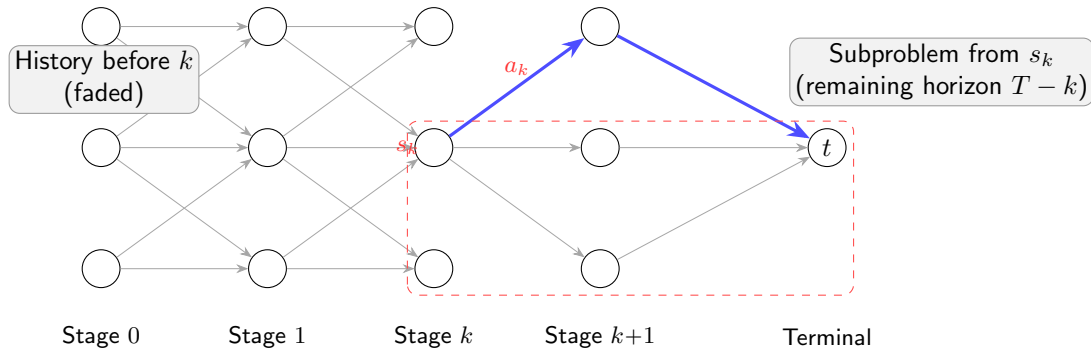
- ▶ directly: reward $r_t(s_t, a_t) = r_t(s_t, \pi(s_t))$
 - ▶ indirectly: state transition $P_t(s'|s, a) = P_t(s'|s, \pi(s))$
- ▶ $V_t^*(s)$: The optimal value function

$$V_t^*(s) = \sup_{\pi} V_t^\pi(s)$$

Principle of optimality and Backward Induction



Principle of optimality and Backward Induction



Principle of Optimality

- ▶ Intuition: the tail of an optimal sequence is optimal for the subproblem that begins where you restart.
- ▶ Consider a policy tail $\pi_{t:}^* = (\pi_t^*, \pi_{t+1}^*, \dots, \pi_{T-1}^*)$ that is optimal from state s at time t . Its first decision rule satisfies

$$\pi_t^*(s) \in \arg \max_{a \in \mathcal{A}_t(s)} \mathbb{E}_{\pi_{t+1}^*} \left[\sum_{\tau=t}^{T-1} \beta^{\tau-t} r_{\tau}(s_{\tau}, a_{\tau}) + \beta^{T-t} R_T(s_T) \mid s_t = s, a_t = a \right],$$

where for $\tau > t$ we take $a_{\tau} = \pi_{\tau}^*(s_{\tau})$.

- ▶ Then for any $t' \in \{t, \dots, T-1\}$ and any state s' reachable at t' under $\pi_{t:}^*$, the continuation rule is optimal for the subproblem:

$$\pi_{t'}^*(s') \in \arg \max_{a \in \mathcal{A}_{t'}(s')} \mathbb{E}_{\pi_{t'+1}^*} \left[\sum_{\tau=t'}^{T-1} \beta^{\tau-t'} r_{\tau}(s_{\tau}, a_{\tau}) + \beta^{T-t'} R_T(s_T) \mid s_{t'} = s', a_{t'} = a \right].$$

Bellman optimality (finite horizon, time-varying)

- ▶ Setup: periods $t = 0, \dots, T$; discount $\beta \in (0, 1)$; time-varying $\mathcal{A}_t(s)$, $r_t(s, a)$, transition $P_t(\cdot \mid s, a)$; terminal payoff R_T .
- ▶ Optimal value as a plan problem:

$$V_t(s) = \max_{\pi_t:} \mathbb{E}_{\pi_t:} \left[\sum_{\tau=t}^{T-1} \beta^{\tau-t} r_{\tau}(s_{\tau}, a_{\tau}) + \beta^{T-t} R_T(s_T) \mid s_t = s \right], \quad V_T(s) = R_T(s).$$

- ▶ By the principle of optimality, one-step decomposition:

$$V_t(s) = \max_{a \in \mathcal{A}_t(s)} \left\{ r_t(s, a) + \beta \mathbb{E}[V_{t+1}(S_{t+1}) \mid s, a] \right\}.$$

- ▶ Optimal decision rule at date t :

$$\pi_t^*(s) \in \arg \max_{a \in \mathcal{A}_t(s)} \left\{ r_t(s, a) + \beta \mathbb{E}[V_{t+1}(S_{t+1}) \mid s, a] \right\}.$$

Backward induction (finite horizon): pseudo-code

- ▶ Step 0 (inputs): horizon T , discount $\beta \in (0, 1)$; primitives $\{\mathcal{A}_t(s), r_t(s, a), P_t(\cdot \mid s, a)\}_{t=0}^{T-1}$; terminal payoff R_T .
- ▶ Step 1 (initialize terminal value): for all states s , set $V_T(s) \leftarrow R_T(s)$.
- ▶ Step 2 (backward sweep): for $t = T - 1, T - 2, \dots, 0$ and each state s ,

$$Q_t(s, a) \leftarrow r_t(s, a) + \beta \mathbb{E}[V_{t+1}(S_{t+1}) \mid s, a] \quad \text{for all } a \in \mathcal{A}_t(s),$$

$$V_t(s) \leftarrow \max_{a \in \mathcal{A}_t(s)} Q_t(s, a), \quad \pi_t^*(s) \in \arg \max_{a \in \mathcal{A}_t(s)} Q_t(s, a).$$

- ▶ Step 3 (outputs): optimal value sequence $(V_t)_{t=0}^T$ and policy sequence $(\pi_t^*)_{t=0}^{T-1}$.
- ▶ Remarks
 - ▶ dynamic programming (DP) decomposes the multi-stage problem via the principle of optimality
 - ▶ the $\arg \max$ at each t is greedy w.r.t. Q_t (globally optimal because V_{t+1} is optimal)

From finite-horizon recursion to the infinite-horizon equation

- ▶ Finite-horizon, time-varying recursion ($t = T - 1, \dots, 0$):

$$V_T(s) = R_T(s), \quad V_t(s) = \max_{a \in \mathcal{A}_t(s)} \{ r_t(s, a) + \beta \mathbb{E}[V_{t+1}(S_{t+1}) \mid s, a] \}.$$

- ▶ Time-homogeneous primitives and infinite horizon:

$$\mathcal{A}_t \equiv \mathcal{A}, \quad r_t \equiv r, \quad P_t \equiv P, \quad \beta \in (0, 1), \quad r \text{ bounded}.$$

- ▶ Stationary infinite-horizon objective:

$$V(s) = \sup_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \beta^t r(S_t, \pi(S_t)) \mid S_0 = s \right].$$

- ▶ Bellman equation (time-homogeneous form):

$$V(s) = \max_{a \in \mathcal{A}(s)} \{ r(s, a) + \beta \mathbb{E}[V(S') \mid s, a] \}.$$

Bellman operators and fixed points

- ▶ Optimal Bellman operator $T : \mathcal{B}(\mathcal{S}) \rightarrow \mathcal{B}(\mathcal{S})$:

$$(TV)(s) = \max_{a \in \mathcal{A}(s)} \{ r(s, a) + \beta \mathbb{E}[V(S') \mid s, a] \}.$$

- ▶ Policy Bellman operator for a fixed stationary policy π :

$$(T^\pi V)(s) = r(s, \pi(s)) + \beta \mathbb{E}[V(S') \mid s, \pi(s)].$$

- ▶ Fixed points:

$$V^* = TV^* \quad (\text{optimal value}), \quad V^\pi = T^\pi V^\pi \quad (\text{value under policy } \pi).$$

- ▶ Greedy policy with respect to a value V (via the one-step value Q_V):

$$Q_V(s, a) = r(s, a) + \beta \mathbb{E}[V(S') \mid s, a], \quad \pi_V(s) \in \arg \max_a Q_V(s, a).$$

The choice is greedy with respect to Q_V ; when $V = V^*$ this yields an optimal policy $\pi^* = \pi_{V^*}$.

Contraction mapping theorem and its consequences

- ▶ Sup norm $\|V\|_\infty = \sup_s |V(s)|$. With bounded rewards and $\beta \in (0, 1)$:

$$\|TV - TW\|_\infty \leq \beta \|V - W\|_\infty, \quad \|T^\pi V - T^\pi W\|_\infty \leq \beta \|V - W\|_\infty.$$

- ▶ Hence T and T^π are β -contractions on $(\mathcal{B}(\mathcal{S}), \|\cdot\|_\infty)$.

- ▶ Consequences:

- ▶ Unique fixed points V^* and V^π .
- ▶ Iteration converges geometrically from any start $V^{(0)}$:

$$V^{(k+1)} \leftarrow TV^{(k)} \rightarrow V^*, \quad V^{(k+1)} \leftarrow T^\pi V^{(k)} \rightarrow V^\pi.$$

- ▶ Error bound after k value-iteration steps:

$$\|V^{(k)} - V^*\|_\infty \leq \frac{\beta^k}{1 - \beta} \|TV^{(0)} - V^{(0)}\|_\infty.$$

Value Function Iteration (time-homogeneous, infinite horizon)

- ▶ Step 0 (inputs): state space $\mathcal{S}$; actions $\mathcal{A}(s)$; reward $r(s, a)$; transition $P(\cdot \mid s, a)$; discount $\beta \in (0, 1)$; tolerance $\varepsilon > 0$.

- ▶ Step 1 (Bellman operator): for any $V : \mathcal{S} \rightarrow \mathbb{R}$,

$$(TV)(s) = \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \mathbb{E}[V(S') \mid s, a] \right\}.$$

- ▶ Step 2 (initialize value): choose $V^{(0)}$ (e.g., $V^{(0)} \equiv 0$); set $k \leftarrow 0$.
- ▶ Step 3 (value update): for each state s ,

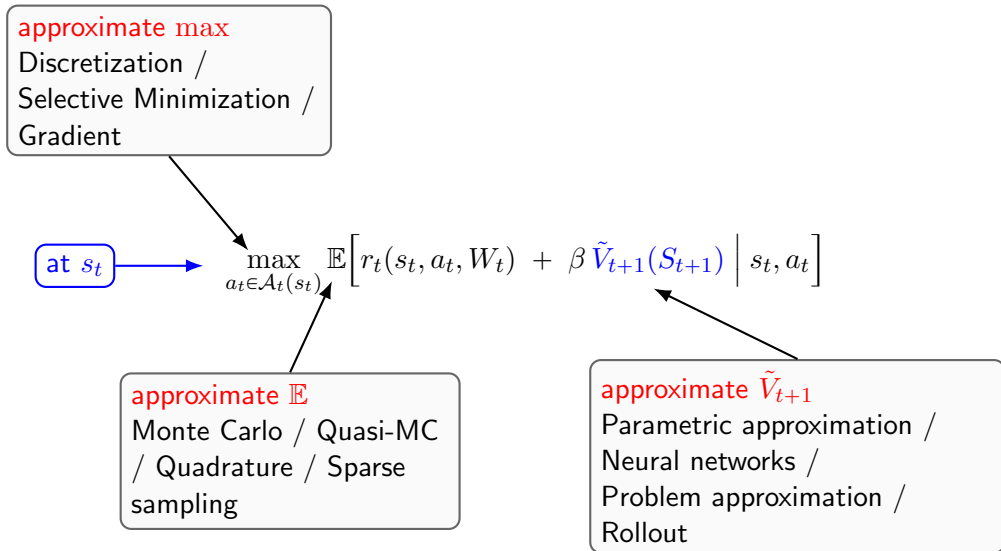
$$Q_{V^{(k)}}(s, a) = r(s, a) + \beta \mathbb{E}[V^{(k)}(S') \mid s, a] \quad \text{for all } a \in \mathcal{A}(s),$$

$$V^{(k+1)}(s) \leftarrow \max_{a \in \mathcal{A}(s)} Q_{V^{(k)}}(s, a).$$

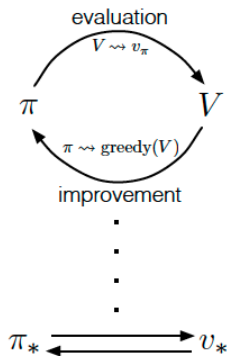
- ▶ Step 4 (convergence check): compute $\Delta_{k+1} = \|V^{(k+1)} - V^{(k)}\|_{\infty}$. If $\Delta_{k+1} \leq \varepsilon$, go to Step 5; else set $k \leftarrow k + 1$ and return to Step 3.
- ▶ Step 5 (policy extraction): greedy w.r.t. $Q_{V^{(k+1)}}$,

$$\pi^*(s) \in \arg \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \mathbb{E}[V^{(k+1)}(S') \mid s, a] \right\}.$$

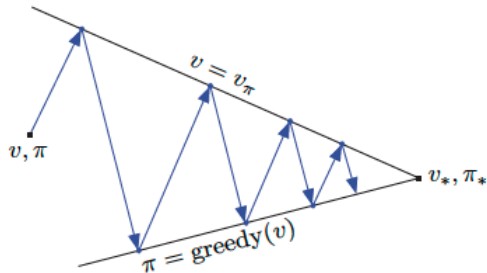
Approximation in Dynamic Programming



Generalized Policy Iteration: two views



(a) View 1



(b) View 2

Figure 1: Generalized Policy Iteration (two views), adapted from [Sutton and Barto \(2018\)](#), Chapter 4.

Decomposing value iteration into evaluation + improvement

Value iteration (one step):

$$V^{(k+1)} = (TV^{(k)})(s) = \max_{a \in \mathcal{A}(s)} \{r(s, a) + \beta \mathbb{E}[V^{(k)}(S') \mid s, a]\}.$$

Equivalent two-step decomposition:

1. improvement (greedy w.r.t. $V^{(k)}$):

$$\pi^{(k)}(s) \in \arg \max_{a \in \mathcal{A}(s)} \{r(s, a) + \beta \mathbb{E}[V^{(k)}(S') \mid s, a]\};$$

2. one-step policy evaluation:

$$V^{(k+1)}(s) = (T^{\pi^{(k)}} V^{(k)})(s) = r(s, \pi^{(k)}(s)) + \beta \mathbb{E}[V^{(k)}(S') \mid s, \pi^{(k)}(s)].$$

Notes: Full evaluation to $V^{\pi^{(k)}}$ yields classical policy iteration; a single backup yields value iteration; in-between gives modified policy iteration. All are instances of GPI.

Generalized Policy Iteration (GPI): idea

- ▶ Maintain a current policy π and a value approximation V (or Q).
- ▶ Two interacting processes:
 - ▶ policy evaluation: push V toward V^π using T^π (exact, iterative, or sample-based);
 - ▶ policy improvement: make π greedier w.r.t. the current V or Q .
- ▶ These can be interleaved, partial, and asynchronous; the combination drives (V, π) toward (V^*, π^*) .
- ▶ With finite MDPs and $\beta \in (0, 1)$, repeated improvement and sufficient evaluation converge to an optimal policy; value iteration and policy iteration are special cases.

Why the decomposition works (sketch)

- ▶ T and T^π are β -contractions in $\|\cdot\|_\infty$ (bounded rewards, $\beta \in (0, 1)$) $\Rightarrow$ unique fixed points V^* and V^π , geometric convergence.
- ▶ Greedy improvement yields π' with $V^{\pi'} \geq V^\pi$ componentwise.
- ▶ Alternating evaluation and improvement produces a monotone sequence bounded by V^* ; sufficient evaluation/exploration $\Rightarrow$ optimality.

References & Further Reading

- ▶ Dimitri P. Bertsekas, **Reinforcement learning and optimal control (2025 Spring course at ASU)**, Lecture 1 and 2:
<https://web.mit.edu/dimitrib/www/RLbook.html>
- ▶ Sutton and Barto, **Reinforcement Learning: An Introduction**, Chapter 4
<https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

Sutton, Richard S. and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.